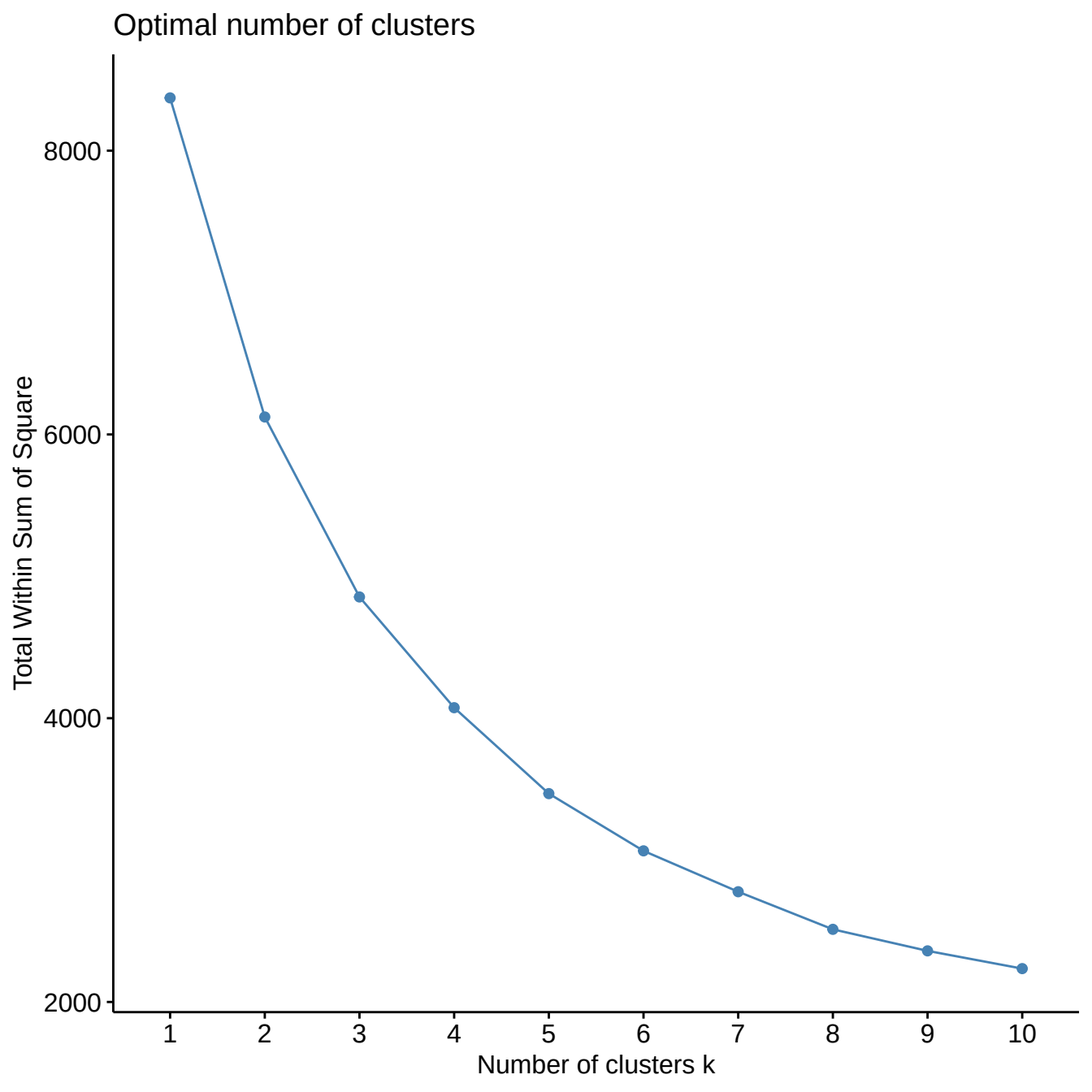
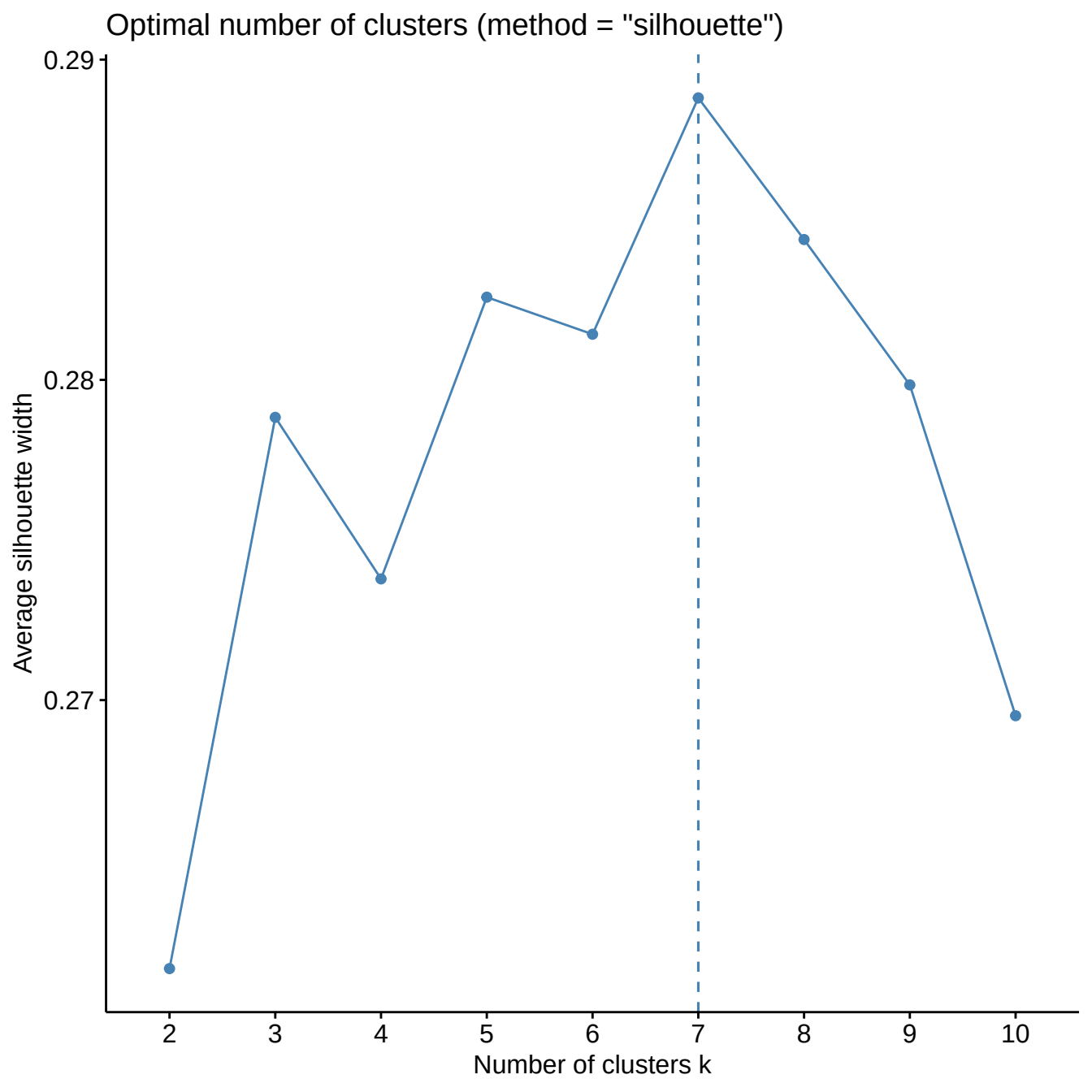
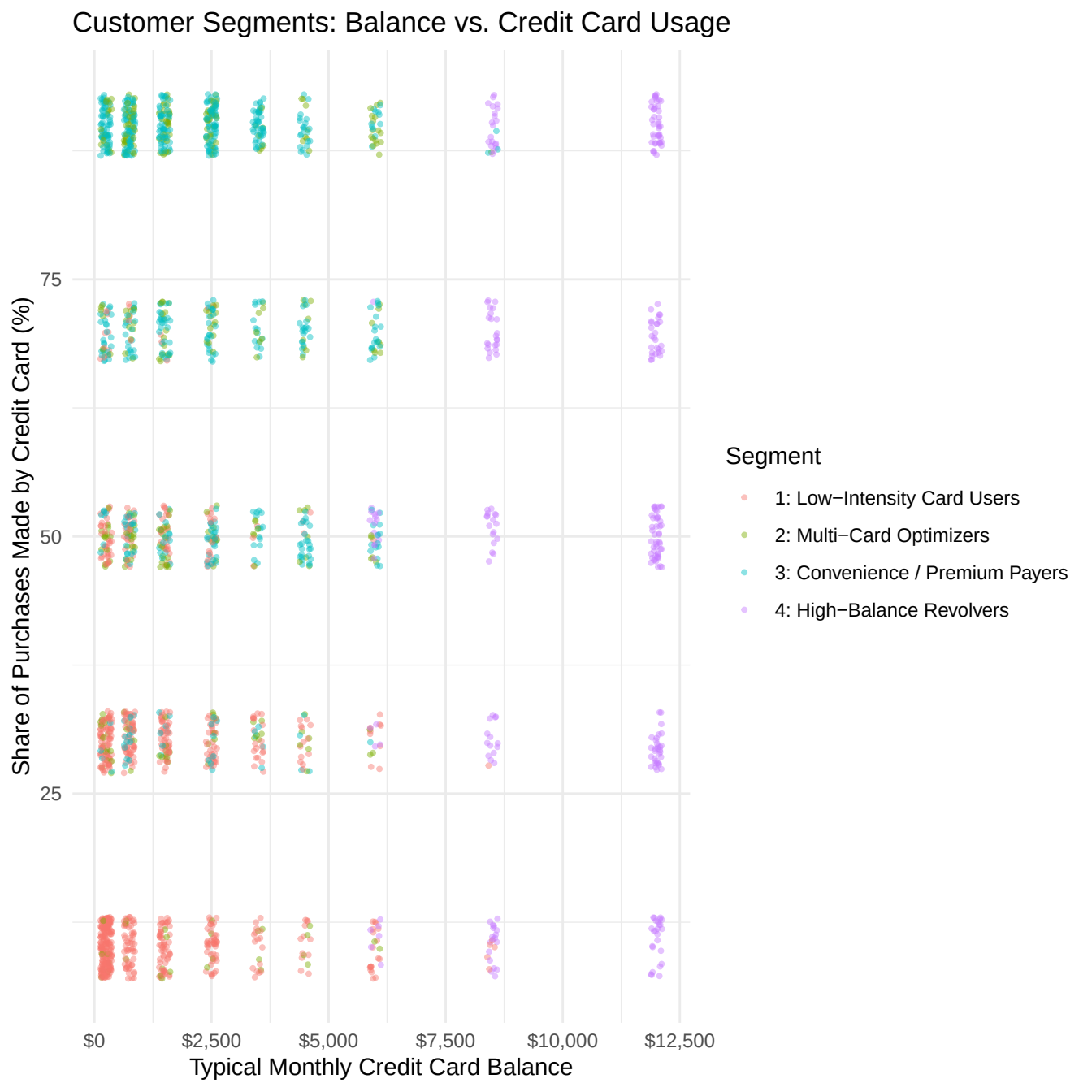
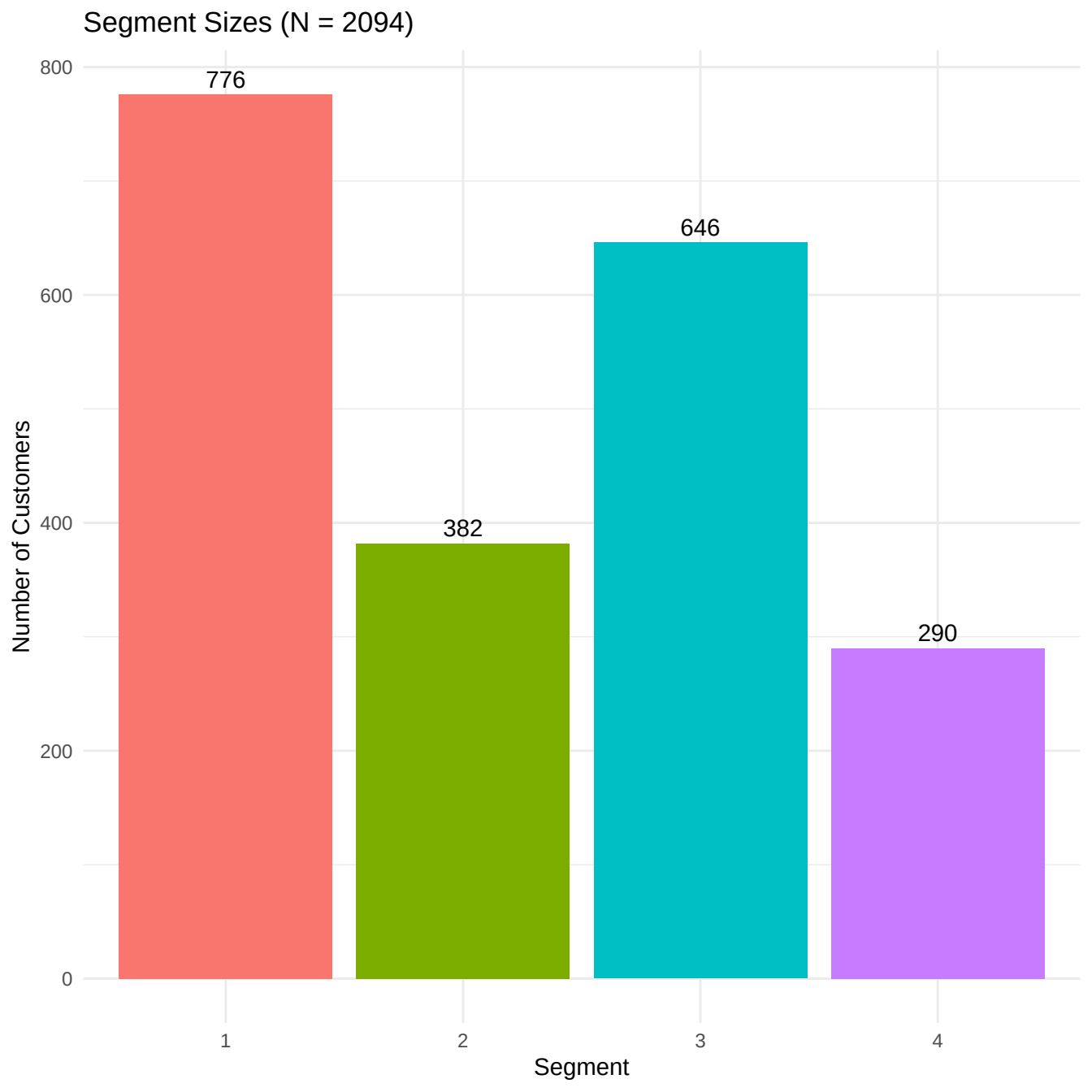
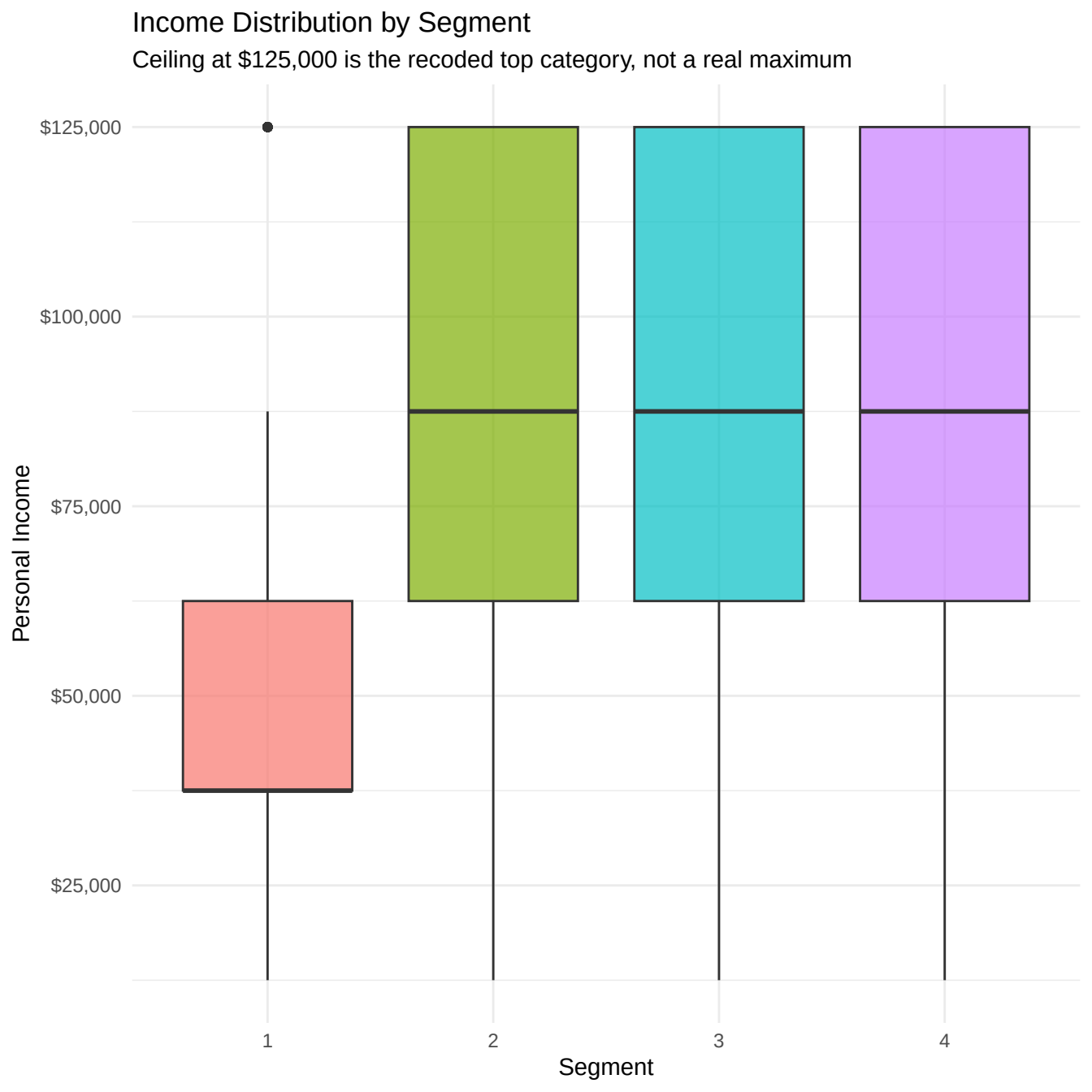
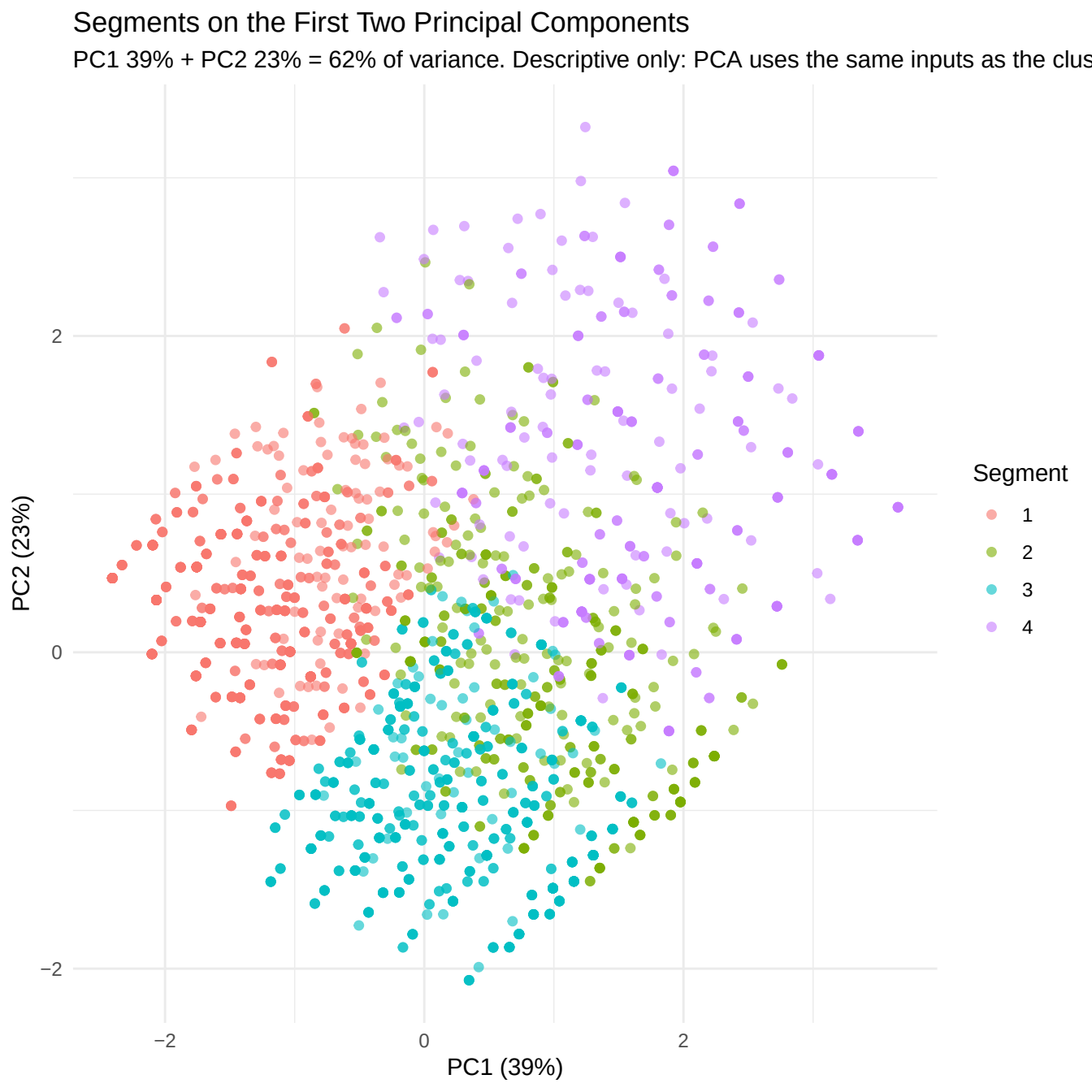

Optimal number of clusters
Total Within Sum of Square
8000
6000
4000
2000
1
2
3
4
5
6
7
8
9
10
Number of clusters k


Optimal number of clusters (method = "silhouette")
0.29
0.28
0.27
Average silhouette width
2
3
4
5
6
7
8
9
10
Number of clusters k


Customer Segments: Balance vs. Credit Card Usage
Share of Purchases Made by Credit Card (%)
75
50
25
$0
$2,500
$5,000
$7,500
$10,000
$12,500
Typical Monthly Credit Card Balance
Segment
1: Low−Intensity Card Users
2: Multi−Card Optimizers
3: Convenience / Premium Payers
4: High−Balance Revolvers


Segment Sizes (N = 2094)
800
600
400
200
0
Number of Customers
776
382
646
290
1
2
3
4
Segment


Income Distribution by Segment
Ceiling at $125,000 is the recoded top category, not a real maximum
$125,000
$100,000
$75,000
$50,000
$25,000
Personal Income
1
2
3
4
Segment


Segments on the First Two Principal Components
PC1 39% + PC2 23% = 62% of variance. Descriptive only: PCA uses the same inputs as the clus
PC2 (23%)
2
0
−2
−2
0
2
PC1 (39%)
Segment
1
2
3
4